ENKU

Base Reader R141

Board engineering review

14 October 2026

NO FAB / NO ASSEMBLY RELEASE

Prepared for preliminary PCBWay review after the planned independent engineering checks. This report records the current board, verified exports and remaining process and product qualification work.

Board	Current specification
PCB	59 x 101mm, four copper layers, nominal1.6mm
Processor	ESP32-S3-WROOM-1-N16R8
Display interface	3.97-inch800x480 e-paper, Base non-touch
Power	BQ24072T + TPS63802, PH-2 battery + separate cell NTC
Verified source	KiCad10.0.7, R141 editable candidate
Digital verification	DRC0 / ERC0 / opens0 / parity0
Physical qualification	NOT RUN

PCB SHA256
b656abcf65c55cdb425a09468ef37918
7b561e764ad1257d2b2761e4f95451fe

Revision and verification scope

R141 is the working PCB. R138 is retained as the comparison baseline. The R139 product qualification and factory-flash documents describe open physical requirements. The legacy r0.1 source basename preserves schematic and project references.

Evidence	Actual result / limitation
Native current R141	DRC0, ERC0, opens0, schematic-to-PCB parity0
Inherited safeguards	20 actual-source checks passed;438 pad definitions across127 references retained
Historical control	27 canonical R138 scopes and20 effective negative cases passed
Independent exports	All four complete copper images and11 Gerbers;371 drill objects including4 plated slots;770 IPC records
Assembly	107 fitted top placements,58 BOM groups;19 BARE_PCB andR67 DNP explicitly classified
Routing graph	30 fast paths resolved;14 fast nets have no centreline length missing both inner ground references
Limits	No measured SI/PI, thermal, EMC, impedance or working-device qualification

Current source and export hashes are retained in the package manifest. Native and independent evidence files are included. All generated documents and column mappings were produced from the unchanged current PCB.

R141 routing changes

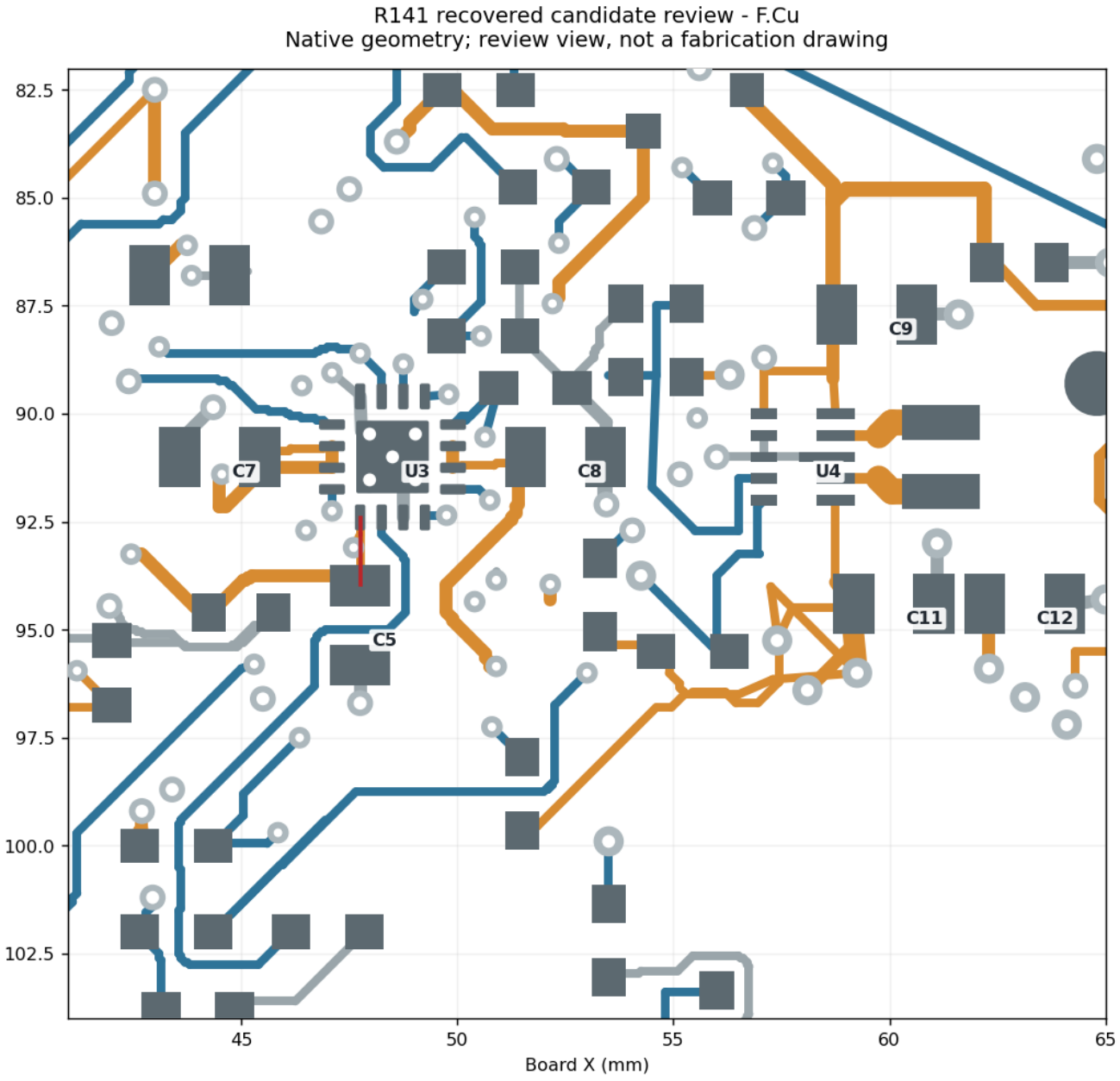
Connection	R138 path (mm)	R141 path (mm)
C5.1 to U3.13 VBUS	29.094	1.550
C7.1 to U3.10 VSYS	16.688	1.12071
C8.1 to U3.2 VBAT	12.504	1.12071
Q1.3 to L2.2	11.308	6.505
Q1.3 to D1.2	26.458	3.144
Q1.3 to C25.1	14.985	3.000

The charger capacitors and EPD switching components were placed closer to their active nodes. Control traces and common/card SPI routing were revised around ground-reference regions. R19 was moved clear of the independently defined ESP32 module body. USB differential copper remains invariant.

Preferred-plane gaps remain on common MOSI0.741064mm and SCLK1.057355mm near MCU. Far-inner ground remains present there. These geometric measurements identify the current routing; they do not establish impedance or timing margins. Ground-via proximity is a review metric, with two transition locations around2.7mm awaiting SI assessment.

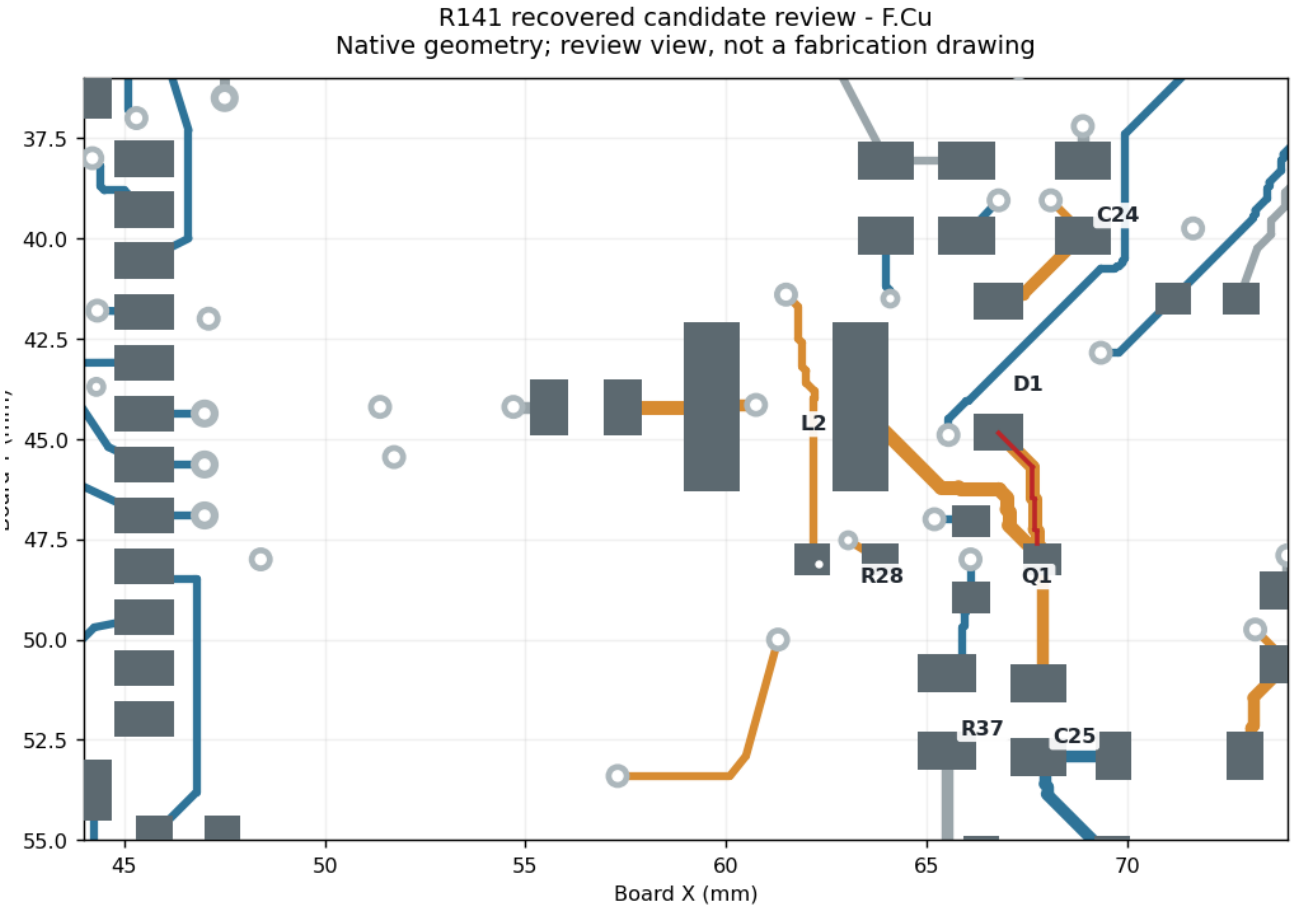
Charger placement and VBUS route

F.Cu view from the current verified board. C5, C7 andC8 provide short local connections to BQ24072T. Ground and supply behaviour still require PI and physical load/fault checks.



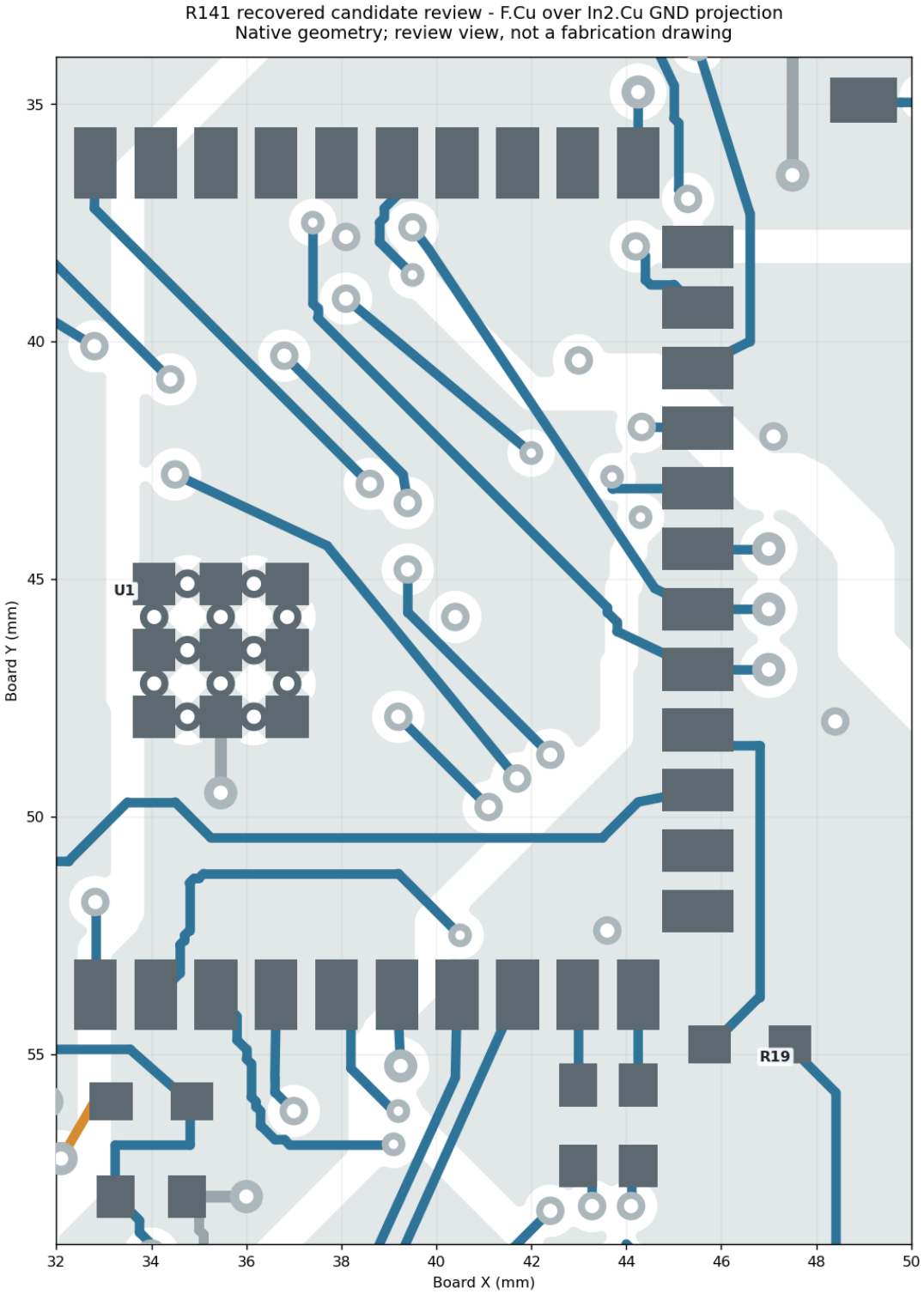
EPD switching area

Current compact switching area. Review switching-current loops, gate/source reference and quiet HV/feedback connections against the exact panel and component requirements.



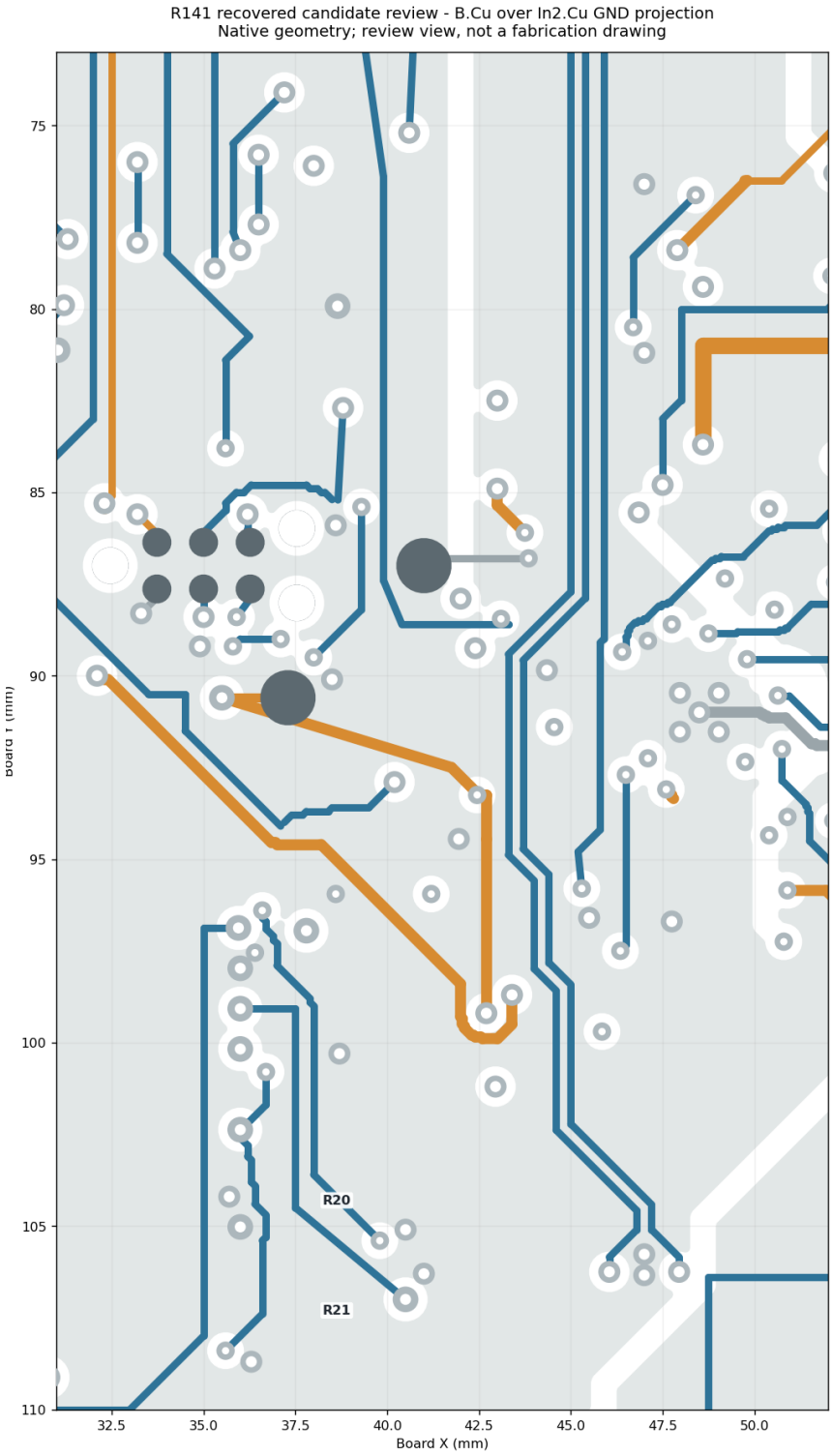
MCU SPI reference region

Current reference overlay. Preferred-plane gaps on common MOSI/SCLK are short and have far-inner ground present. This view is geometric evidence, not a field-solver result.



microSD SPI reference region

Current card routing and ground-reference overlay. USB-MSC sharing the same microSD remains a firmware and hardware qualification requirement.



Fabrication and assembly review

Item	Review instruction
Layer order	F.Cu / In1.Cu / In2.Cu / B.Cu. Actual material, copper and dielectric stack are unapproved.
USB	Review90-ohm differential impedance with actual stack and In2 reference; no silent resizing.
Vias	346 routed vias; nominal ring >=0.15mm. See exact size counts in package_counts_R141.json.
Six fill/cap sites	Five U3 EP andone R28. Resin fill/copper cap, front untented/back tented. Exact process CSVs included.
U1 exposed pad	9-land/9-window grid,12 intrinsic plated0.30mm EP holes. Confirm solder wicking, stencil/reflow and joint inspection.
USB stakes/slots	GCT USB4215-03-A only. Four plated slots,0.8mm stakes on1.6mm PCB; hybrid solder/mechanical process review required.
R67 and bare PCB	Do not populate R67.19 bare references are mechanical/service/NTC/test features.
Supplier acceptance	Submit explicit job-specific DFM/process answers before any production release.

Coordinates, orientation and package use

Manufacturing coordinates use millimetres and a common absolute0,0 origin: X=KiCad board X; Y=-KiCad board Y. Gerber envelope X18..77mm, Y-121..-20mm. Native CPL and PCBWay aliases retain identical coordinates and native angles. All107 placements are top/front.

Supplier library zero-angle, pin1 and polarity have not been normalized or approved. Verify each package using the complete native pad map and assembly drawings. Hybrid connector stakes require their own process review. The first-pad index is a locator aid; pad numbers and manufacturer drawings control orientation.

File group	Contents
Fabrication	11 Gerbers, PTH/NPTH drills, drill maps/report, IPC-D-356 and Gerber-only ZIP
Assembly	Native and mapped BOM/CPL CSV, Excel workbook,127-reference first-pad index,438-pad map and6 via-process requests
Drawings	This report, eight native layer/assembly views and five schematic sheets
Editable/evidence	Current project-local KiCad source/libraries and native/independent verification JSON
Qualification	R139 product/flash requirements plus R141 pre-review methods and closure record

Native board PDF views are auto-fit and are not physical-scale fabrication drawings. Gerber Edge.Cuts controls the actual outline. No validated factory firmware or test binary is supplied.

Checks before sending to PCBWay

Method	Required next evidence
Schematic review / FMEA	Independent pin-by-pin power/startup/reset/charger/NTC/fault review and decision record.
SI / return paths	Review reference continuity, transition returns, SPI timing and USB geometry with stack sensitivity.
PI / thermal	DC/AC supply paths, effective MLCC capacitance, worst-case load/transient/charge power and thermal limits.
Assembly / DFM	MPN-to-footprint/polarity audit, six fill/cap sites, stencil and hybrid connectors, exact package import.
Mechanical	Case, exact display/FPC, antenna keepout, USB/button access, thin protected battery and retained NTC.
Firmware / factory	Startup limits, USB-MSC/microSD ownership, recovery, flash manifest, fixture and numeric FCT limits.
Physical qualification	Real board/panel/battery/NTC, USB power/current, charging/faults, thermal and repeated load/reset tests.

Each finding needs an evidence location, disposition and closure condition. A software PASS cannot close a physical test. The detailed checklist is included as PRE_REVIEW_CHECKS_R141.md.

Product limits and source references

Base uses physical controls, non-touch e-paper, microSD and a hard reader-load switch. No frontlight, Hall cover, dock or Qi is part of this board. Case thickness must not increase. BQ24072T charging is445mA nominal, modeled maximum492.42mA and BAT maximum4.24V. Qualified packs, separate cell NTC retention, exact display/case fit and firmware are still open.

Physical startup, charging, NTC faults, combined Wi-Fi/SD/EPD loads, thermal behaviour, USB-MSC, PSRAM, fixture/FCT and consumer qualification have not been run. See the retained product and factory-flash requirements for the planned groups and holds.

Project source and evidence: engineering/r141-routing-power-fixes, ENKU-Base-reader. The source PCB hash on page1 and package SHA256 manifest identify the actual candidate. Historical R138 verification is a regression control.

Official PCBWay assembly file requirements, consulted14October2026: pcbway.com/assembly-file-requirements.html. Included BOM CSV/Excel, fabrication layers, X/Y/rotation/side data and supplementary drawings follow the listed file categories. Approval of supplier import and manufacturing processes remains pending.

Original hardware: CERN-OHL-W2.0. Software: MIT. Original documentation: CC BY4.0. Third-party terms and ENKU trademark exceptions remain as specified in LICENSE.md.

Layer names and hidden objects

Read-only native KiCad 10.0.7 audit of the exact R141 PCB. All 16 enabled layers have names, and all four copper layers are present in the independently verified manufacturing exports.

The disabled User.Drawings layer contains 45 historical drawing/text objects, including obsolete BQ25185 and early placement notes. It is excluded from fabrication. These notes must not be used as current design or process instructions. Cleanup is a next-pass item.

79 component references and 120 values are hidden. Other hidden fields contain component metadata. Full reference/first-pad and complete pad maps remain in the package. Hidden text does not conceal routed copper from native checks or explicit manufacturing exports.

No copper graphics without net semantics, unassigned tracks, unassigned vias or unassigned copper zones were found. The 19 items with no net are two antenna/keepout rule areas, nine NPTH mechanical pads and eight SMD pads on SW1, J1 and J2. Six SMD pads are mechanical solder lands. J2 pads9 and10 are unassigned connector contacts; card-detection and mechanical qualification remain open and require a separate pin-function review.

27 pads use the native all-copper wildcard for plated or mechanical holes. Inactive In3..In30 entries in these pad masks do not add copper layers to this four-layer board. No tracks, vias or zones touch disabled layers.

Detailed inventory: layer_visibility_audit_R141.json and Disabled_legacy_drawings_R141.csv. The PCB hash and all fabrication exports remain unchanged.